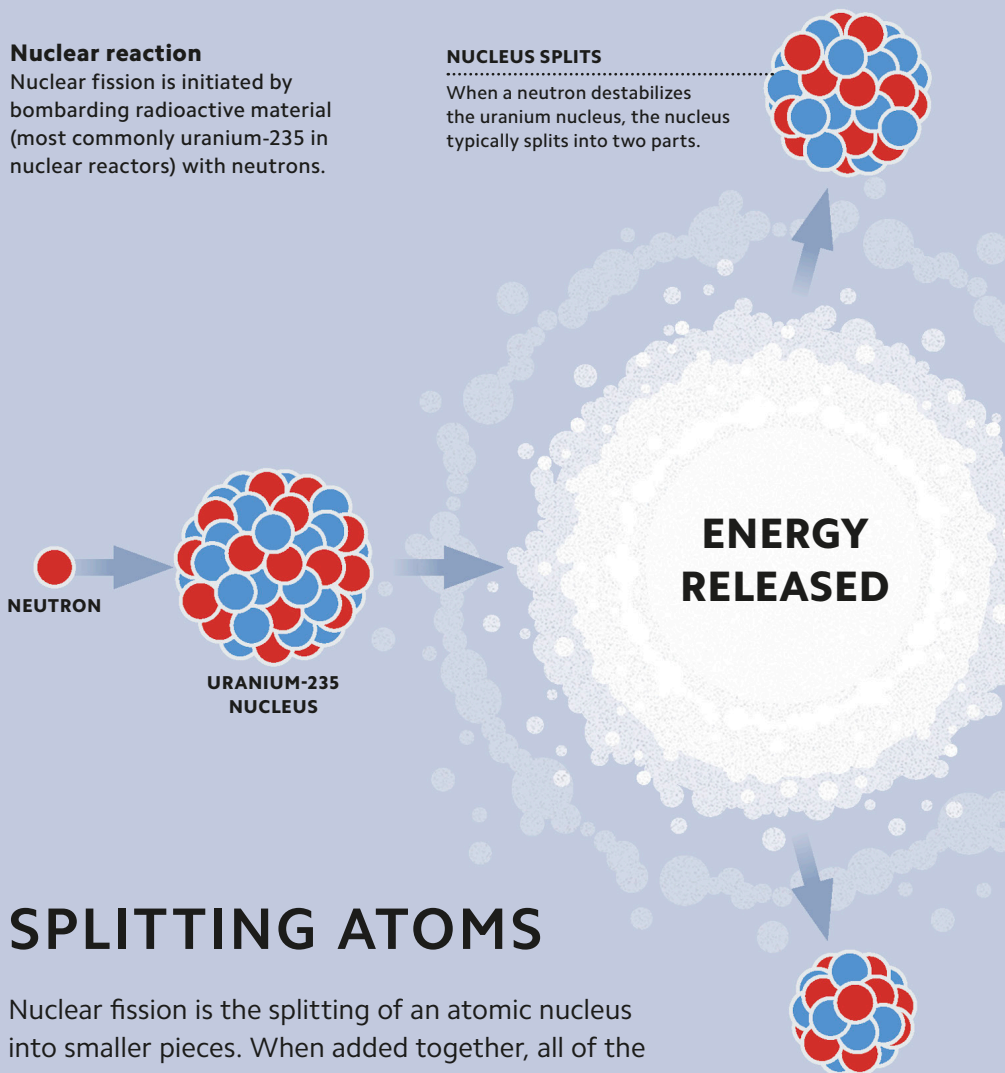


Nuclear reaction

Nuclear fission is initiated by bombarding radioactive material (most commonly uranium-235 in nuclear reactors) with neutrons.

NUCLEUS SPLITS

When a neutron destabilizes the uranium nucleus, the nucleus typically splits into two parts.



SPLITTING ATOMS

Nuclear fission is the splitting of an atomic nucleus into smaller pieces. When added together, all of the lower-mass fragments left after fission weigh less than the original heavy nucleus—the missing mass is released as energy. Neutrons released in fission can go on to collide with other nuclei and can cause them to undergo fission. This is called a chain reaction (such as in a nuclear reactor or an atomic weapon).